

HW3

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compiled with L^AT_EX

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Section 1.6 #37

answer the questions about the linear system

$$\begin{cases} -x & -z = -1 \\ 2x & + 2z = 1 \\ x - 3y - 3z = 1 \end{cases}$$

(a)

$$\begin{array}{l} -1x + 0y - 1z = -1 \\ 2x + 0y + 2z = 1 \\ 1x - 3y - 3z = 1 \end{array} \implies A = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 0 & 2 \\ 1 & -3 & -3 \end{bmatrix}$$

(b)

finding the determinant:

$$\begin{aligned} \det(A) &= -1 \begin{vmatrix} 0 & 2 \\ -3 & -3 \end{vmatrix} - 0 \begin{vmatrix} 2 & 2 \\ 1 & -3 \end{vmatrix} + (-1) \begin{vmatrix} 2 & 0 \\ 1 & -3 \end{vmatrix} \\ &= -1(0 - (-6)) - 0 + (-1)(-6 - 0) = -6 + 6 = 0 \end{aligned}$$

(c)

nope. $\det(A) = 0$, so A is not invertible, and $A\vec{x} = \vec{b}$ does not have a unique solution.

(d)

row reduce the matrix

$$\begin{aligned} & \left[\begin{array}{ccc|c} -1 & 0 & -1 & -1 \\ 2 & 0 & 2 & 1 \\ 1 & -3 & -3 & 1 \end{array} \right] \xrightarrow{(-1)R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 2 & 0 & 2 & 1 \\ 1 & -3 & -3 & 1 \end{array} \right] \\ & \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & -3 & -4 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & -3 & -4 & 0 \\ 0 & 0 & 0 & -1 \end{array} \right] \end{aligned}$$

row 3 says $0x + 0y + 0z = -1$, i.e. $0 = -1$, which is impossible. so it has no solutions.

Section 1.7 #25

find A^{-1} using an LU factorization, where

$$A = \begin{bmatrix} 1 & 4 \\ -3 & -11 \end{bmatrix}$$

row reduce A to upper triangular form:

$$\begin{bmatrix} 1 & 4 \\ -3 & -11 \end{bmatrix} \xrightarrow{R_2 + 3R_1} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = U, \quad L = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

we can check:

$$LU = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ -3 & -11 \end{bmatrix} = A \quad \checkmark$$

invert L and U :

$$L^{-1} = \frac{1}{(1)(1) - (0)(-3)} \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \quad U^{-1} = \frac{1}{(1)(1) - (4)(0)} \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

since $A = LU$, $A^{-1} = U^{-1}L^{-1}$:

$$A^{-1} = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} (1)(1) + (-4)(3) & (1)(0) + (-4)(1) \\ (0)(1) + (1)(3) & (0)(0) + (1)(1) \end{bmatrix} = \begin{bmatrix} -11 & -4 \\ 3 & 1 \end{bmatrix}$$

and then check:

$$AA^{-1} = \begin{bmatrix} 1 & 4 \\ -3 & -11 \end{bmatrix} \begin{bmatrix} -11 & -4 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} -11 + 12 & -4 + 4 \\ 33 - 33 & 12 - 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \checkmark$$